ENAE464 - 0202

Lab05: Ludwieg-Tube Experiments on a Free-Flying Cone

Due on April 17th, 2026 at 11:59 PM

Dr. Silbaugh, 02:00 PM

Mikołaj Kostrzewa & Vai Srivastava

Experiment Performed: April 10th, 2026
Report Submitted: April 15th, 2026

April 15th, 2026

Enclosed is the technical report for the *Ludwieg-Tube Experiments on a Free-Flying Cone* laboratory experiment. This report presents the experimental methodology, results, and analysis of flow characteristics and effects for a free-flying cone in a Mach-4 Ludwieg tube.
Please feel free to contact us with any questions regarding the contents of this report.

Respectfully,
Mikołaj Kostrzewa & Vai Srivastava

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1 Abstract

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2 Introduction

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2.1 Theoretical Background

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2.2 Objectives

The specific objectives of this experiment are:

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3 Methodology

3.1 Experimental Apparatus

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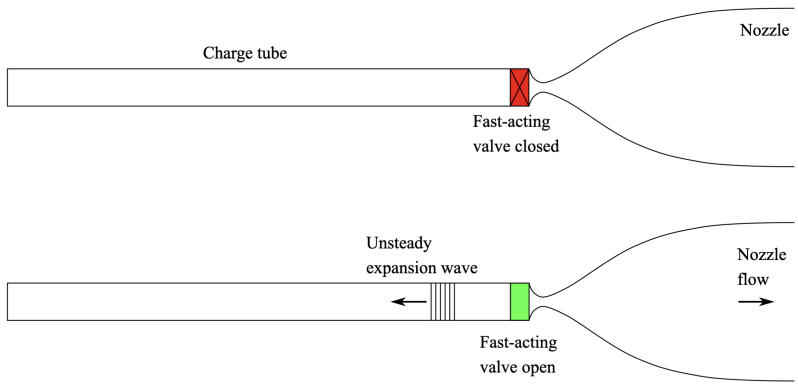

Figure 3.1.1: Schematic diagram of a Ludwieg tube [1].

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3.2 Operating Procedure

The experimental procedure was as follows:

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4 Results

4.1 Freestream Flow Conditions

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4.2 Drag

4.2.1 Theoretical

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4.2.2 Experimental

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4.3 Shock Angle

4.3.1 Theoretical

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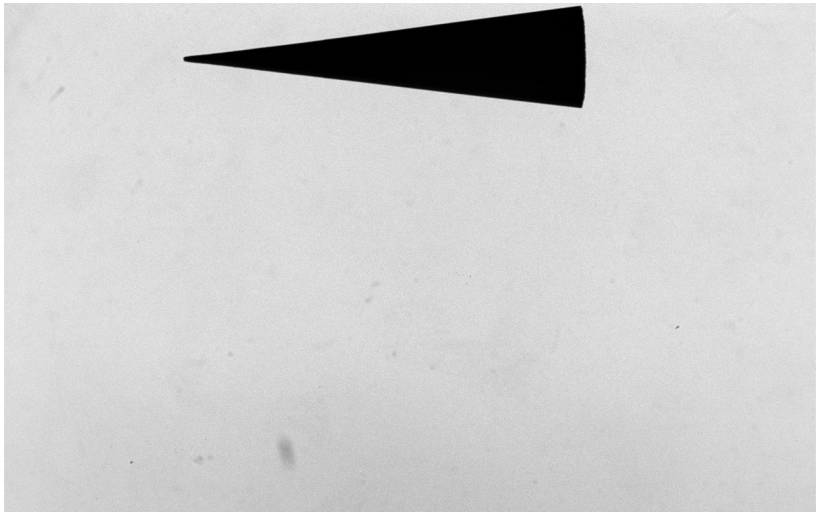

Figure 4.3.1: Theoretical Shock Shape across the Free-Flying Cone Body.

4.3.2 Experimental

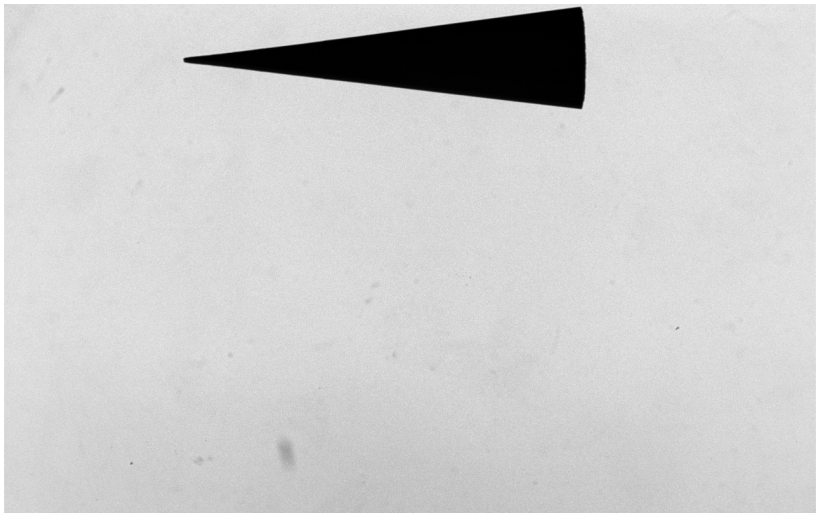

Figure 4.3.2: Experimental Shock Shape across the Free-Flying Cone Body.

5 Analysis and Discussion

5.1 Drag

5.1.1 Comparison with Theory

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5.1.2 Viscous Component

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5.2 Shock Angle

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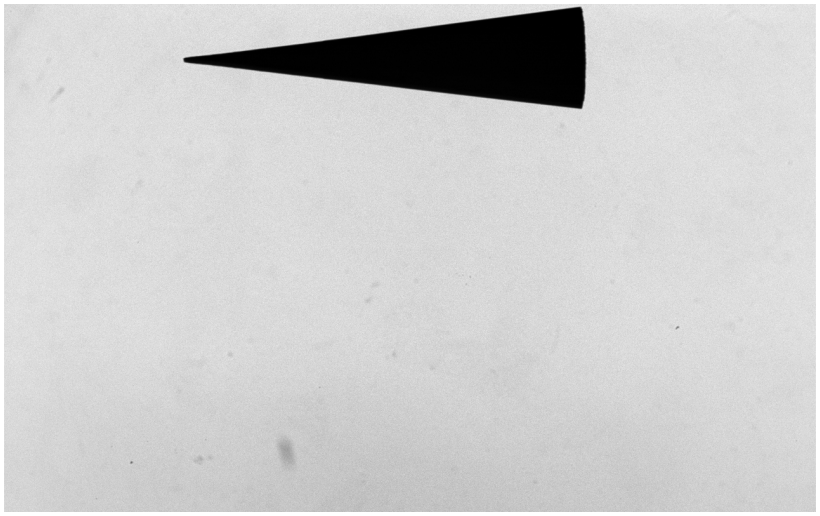

Figure 5.2.1: Comparison of Theoretical (Red) and Experimental (Blue) Shock Shape across the Free-Flying Cone Body.

6 Summary and Conclusions

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7 References

[1] S. J. Laurence and R. S. de Silva, *Ludwieg-tube experiments on a free-flying cone*, University of Maryland, April 8, 2026. [Online]. Available: https://github.com/vaisriv/enae464-lab05/blob/main/references/Ludwieg_tube_cone.pdf (see p. 2).

[2] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, ISBN: 9781266076442.

[3] S. J. Laurence, *Matlab cone*, April 13, 2026. [Online]. Available: https://github.com/vaisriv/enae464-lab05/blob/main/lib/matlab_cone.

8 Appendices

8.1 Data

For the sake of brevity, the raw measurement data will not be displayed in this report. However, in the spirit of completeness, the repository containing the complete data, source code, and notes for this report can be found at github:vaisriv/enae464-lab05.

8.2 Code

Similarly, only the code files that are key to the analysis are included below, and can be accessed in full in the aforementioned GitHub repository.

Listing 8.2.1: Data Analysis Script.
`./src/index.py`

```python
"""
ENAE464 Lab05: Ludwieg-Tube Experiments on a Free-Flying Cone
Analysis of a free-flying cone in a Mach-4 Ludwieg tube
"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
from pathlib import Path
from scipy.optimize import curve_fit
```

```python

# --- Paths ------------------------------------------------------------------

DATA_DIR = Path("./data/mfg_m4_cone")
OUTPUT_TEXT_DIR = Path("./outputs/text")
OUTPUT_FIGS_DIR = Path("./outputs/figures")

OUTPUT_TEXT_DIR.mkdir(parents=True, exist_ok=True)
OUTPUT_FIGS_DIR.mkdir(parents=True, exist_ok=True)

# --- Header -----------------------------------------------------------------

print("=" * 80)
print("ENAE464 Lab05: Ludwieg-Tube Experiments on a Free-Flying Cone")
print("=" * 80)
print()

# --- Constants --------------------------------------------------------------

# FIXME:

# --- Analysis ---------------------------------------------------------------

# FIXME:

# --- Visualization ----------------------------------------------------------

# FIXME:

# --- Output -----------------------------------------------------------------

# FIXME:
```